QUANT

Averages, LCM & Decimals

Assignment 01

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1. Which of the following fractions is the largest?

1. $\frac{12}{29}$ 2. $\frac{108}{129}$ 3. $\frac{76}{130}$ 4. $\frac{101}{200}$

2. Of 120 people in a room, $3/5$ are women. If $2/3$ of the people are married, what is the maximum number of women in the room, who could be unmarried?

1. 80 2. 72 3. 48 4. 40

3. The Plush Hotel received a shipment of glasses packed in full cartons of 40 glasses each. Another shipment of glasses packed 24 to a carton went to the Maison Restaurant. Glasses were also shipped to State University, but this shipment contained cartons with 25 glasses each. If the hotel, the restaurant and the university each received the same number of glasses and if none of them received more than a thousand glasses, how many cartons were shipped totally?

1. 64 2. 480 3. 600 4. 1200

4. $\sqrt{1-\left(\frac{2}{9}+\frac{1}{36}+\frac{1}{18}\right)}=?$

1. $1/5$ 2. $\sqrt{2/3}$ 3. $5/6$ 4. $\sqrt{3}$

5. What is the lowest number that is divisible by the numbers 1 through 10?

1. 1,000 2. 2,520 3. 1,200 4. 5,040

6. In the SY B.Com class of 360 students, 5/12 are women. In the FY B.Com class of 350 students 5/7 are women. How many more women are there in the FY B.Com class than in the SY B.Com class?

1. 250 2. 150 3. 100 4. 50

7. The price of Bull's Eye (also a kind of sweet) increases from 5 for Rs. 7 to 3 for Rs. 7. Compared to earlier, how many bull's eye less will Rs. 3.5 fetch you now?

1. $2/7$ 2. $52/35$ 3. 1 4. 2

8. Compute x if: $x = \frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\frac{1}{4.5}+\frac{1}{5.6}+\frac{1}{6.7}+\ldots\frac{1}{(n-1)n}+\frac{1}{n(n+1)}$

1. $\frac{n}{n-1}$ 2. $\frac{2n}{n+1}$ 3. $\frac{n}{n+1}$ 4. 2n

9. In a parking lot, every third car is red and every fourth car is white. What could be the maximum number of cars in that parking lot?

1. 12 2. 16 3. 13 4. 11

10. One-fourth of my marks in English is equal to one third of my marks in Hindi, The total number of marks secured by me in both the subjects is 140. The marks secured by me in English are

1. 60 2. 80 3. 75 4. None of these

Answer Key & Explanations

Test – 01	
1.	2
2.	4
3.	1
4.	3
5.	2
6.	3
7.	3
8.	3
9.	4
10.	2

Test - 01

1. I option is less than 0.5, III and IV options are very close but a bit more than 0.5, II option given is much bigger than 0.5. Therefore this should be the largest.

2. No. of women = $120 \times \frac{3}{5} = 72 \Rightarrow$ men = 48,

Married people = $\frac{2}{3} \times 120 = 80 \Rightarrow$ Unmarried = 40

To find maximum no. of unmarried women, assume that all men are married. That is, out of 80, 48 are men. Hence 32 will be the number of married women. Hence maximum number of unmarried women = 40

3. If these three places received the same number of glasses, call the number N, the N must be a multiple of 40, of 24, and of 25. In this situation, N is called a common multiple of 40, 24, and 25. You are asked to find the least common multiple. N = 600. To find the number of cartons in each shipment, you need to divide 600 (the total number of glasses shipped to each location) by the number of glasses per carton shipped to that particular location. So, for instance, the number of cartons shipped to the Plush hotel would be $\frac{600}{4} = 15$ cartons. Similarly the other two places would have 24 and 25 cartons. A total of 24 + 25 + 15 = 64.

4. $\left(\frac{2}{9}+\frac{1}{36}+\frac{1}{18}\right) = \frac{11}{36}$. And $1-\frac{11}{36} = \frac{25}{36}$,

and its square root is $\frac{5}{6}$.

5. You are asking for the Least Common Multiple of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10.
To solve such problems, you can write the prime-power factorization of each of the numbers, and the exponent of the prime powers that occur:

	exponent of number factorization	2	3	5	7
1	1	0	0	0	0
2	2^1	1	0	0	0
3	3^1	0	1	0	0
4	2^2	2	0	0	0
5	5^1	0	0	1	0
6	$2^1 \times 3^1$	1	1	0	0
7	7^1	0	0	0	1
8	2^3	3	0	0	0
9	3^2	0	2	0	0
10	$2^1 \times 5^1$	1	0	1	0

Now take the maximum in each column, and use those as exponents for your number:
$2^3 \times 3^2 \times 5^1 \times 7^1 = 2{,}520$.

6. No. of women in SYBcom class = $\frac{5}{12} \times 360 = 150$,

No. of women in FYBcom class = $\frac{5}{7} \times 350 = 250$

Difference = 250 – 150 = 100

7. No. of bulls eye that can be purchased for Rs. 3.5 at old price = $\frac{5 \times 3.5}{7} = 2.5$

No. of bulls eye that can be purchased for Rs. 3.5 at new price = $\frac{3}{7} \times 3.5 = 1.5$. Difference = 2.5 - 1.5 = 1.

8. : $x = (\frac{1}{1} - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \ldots..(\frac{1}{n} - \frac{1}{n+1})$ x

$= 1 + 0 + 0 + 0 + 0 + \ldots\ldots\ldots 0 + 0 + \frac{-1}{n+1}$

x = 1 - $\frac{1}{n+1} = \frac{n}{n+1}$. So 3rd option.

9. Every 3rd car is red and every fourth car is white.
On the face of it seems that the data is inadequate to answer this question. But take the LCM of 3 and 4 i.e. 12. $\Rightarrow$ 12th car is red as well as white, which can't be true. The maximum number of cars in parking lot is 11.

10. Let marks in English = x & Hindi = y. $\frac{x}{4} = \frac{y}{3}$

Also x + y = 140. Solving the 2 equations we get x = 80